CTO CAPSTONE PROJECT · 2026

CTO360

Enterprise Technology Control Tower

One View. Connected Decisions. Measurable Technology Outcomes.
Moving an enterprise from current state to target state, measurably, with governed AI.



7 source systems · 3 business units · 1 decision layer

Ram
CTO

Suman
Strategy & ROI

Vaibhav
Regulatory & AI Governance

Santhosh
Finance & Investment

Pankaj
Operations

AGENDA

What We Will Cover

About 25 minutes of story and live demo, followed by questions.

01

The Problem

Why CTOs cannot steer AI-led transformation today

02

The Solution

CTO360: a decision layer over existing tools

03

How It Works

Architecture, simulation and how we measure change

04

Live Demo

Five owner workspaces and one connected decision

05

Frameworks & Value

Frameworks applied with metrics, value case, roadmap

06

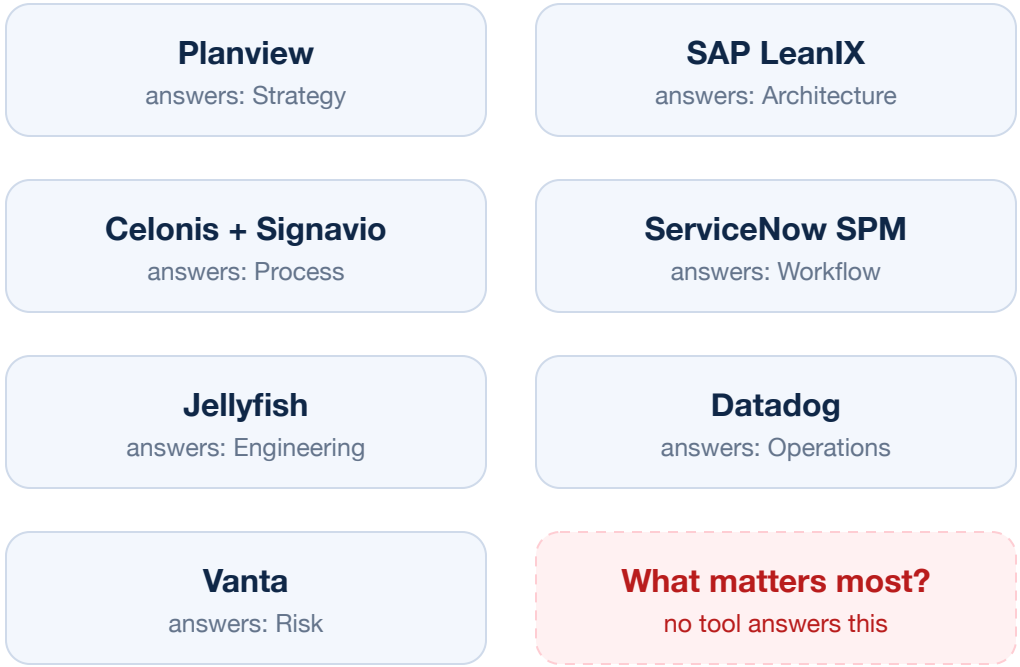
Q&A

Questions and discussion

A Decision Problem, Not a Data Problem

The CTO cannot see across the enterprise fast enough to decide where to intervene, or prove that technology spend produced business outcomes.

- **Fragmented signals.** Strategy, architecture, process, delivery, operations and risk each live in a separate tool, and each answers only its own question.
- **Cause and effect span tools.** A payments programme slips because of legacy dependencies (LeanIX), a team capacity gap (Jellyfish) and rising incidents (Datadog). No single tool shows that chain, so it is found late.
- **Reporting is not deciding.** Dashboards and status decks are plentiful, but signal to decision to action to outcome is manual, slow and rarely closed.
- **Technology spend is hard to defend.** Value realised and risk avoided are not tied to specific decisions.
- **Every business unit reports differently,** so the CTO cannot compare them or move money and people with confidence.



01 · THE PROBLEM

Same Problem, Three Different Realities

Three simulated enterprises with different industries, but the same root cause: signals that are never connected into a decision.

BANKING

Meridian Bank

Business problem: Legacy core and integration layer slowing digital payments; technology debt behind P1 incidents.

WHAT IT COSTS

- Q4 payments milestone at risk
- ₹38.9 Cr value at stake
- Regulatory exposure (RBI, DPDP)

MANUFACTURING

Arvant Industries

Business problem: MES and plant system instability causing line stops; smart factory programme delayed.

WHAT IT COSTS

- Lost production hours
- Late deliveries to customers
- OEE target missed

RETAIL

Orbit Retail

Business problem: Festive peak readiness and inventory sync; engineering capacity stretched.

WHAT IT COSTS

- Lost festive sales
- Overselling and stockouts
- Point-of-sale security risk

The pattern repeats across industries, so the answer must be built once and configured per industry. Retail was added by configuration only.

01 · THE PROBLEM

Our Capstone Problem Statement

Five verbs, five owners, one system. Each owner takes one real business problem (BP1) from current state to target state.

Enterprises cannot see, govern, fund and run AI-led transformation as one system. CTO360 simulates a real enterprise and moves each area from its current state to a target state, measurably.

See

One comparable view across business units, sources and functions

OWNER · RAM

Govern

Regulation, AI models and data under control with evidence

OWNER · VAIBHAV

Fund

Every rupee traceable to a service, with value proven after spend

OWNER · SANTHOSH

Run

Root causes fixed and capacity ready before the peak

OWNER · PANKAJ

Transform

Any organisation moved from AI pilots to measurable value

OWNER · SUMAN

3
simulated enterprises: banking, manufacturing, retail

30
transformation measures, 6 per owner, baseline to target

6
golden-thread steps connecting all five owners live

1
decision layer: AI recommends, humans decide

Team, Ownership and Business Problems

Each owner has a dedicated live workspace in CTO360. Clear splits remove overlap between roles.

R — Ram

CTO · CTO Control Tower

- Multi-domain framework
- Decision intelligence and agents

S — Suman

Strategy & ROI · AI Transformation

- Identify ROI before investment
- Transform any organisation to AI

V — Vaibhav

Regulatory & AI Governance

- Automate regulatory compliance (Banking)
- DPDP and AI governance

S — Santhosh

Finance & Investment Governance

- Governance system for investment
- CAPEX and OPEX control

P — Pankaj

Operations · ERP RCA & Capacity

- ERP root-cause analysis
- Infrastructure capacity planning
- Production automation

- **Suman identifies ROI before spend;** Santhosh proves value after spend.
- **Pankaj says how much capacity is needed and when;** Santhosh decides how to fund it.

- **Investment governance (Santhosh)** is different from regulatory and AI governance (Vaibhav).
- **One RACI matrix** on the Team page is the single source for who is responsible, accountable, consulted and informed.

02 · THE SOLUTION

CTO360: A Decision Layer, Not Another Tool

CTO360 sits on top of the tools an enterprise already owns and turns their signals into decisions that are owned, acted on and measured.



- **One view:** 3 business units × 7 source systems × 16 enterprise functions, all scored the same way so they can be compared.
- **Correlation:** records are linked across tools through shared IDs, so every decision card shows why it exists, with traceable evidence.
- **Prioritised Top 10:** each decision shows impact, money asked versus value protected, confidence, owner and deadline.
- **Closed loop:** approve a decision, issue actions, and track the outcome against baseline and target.

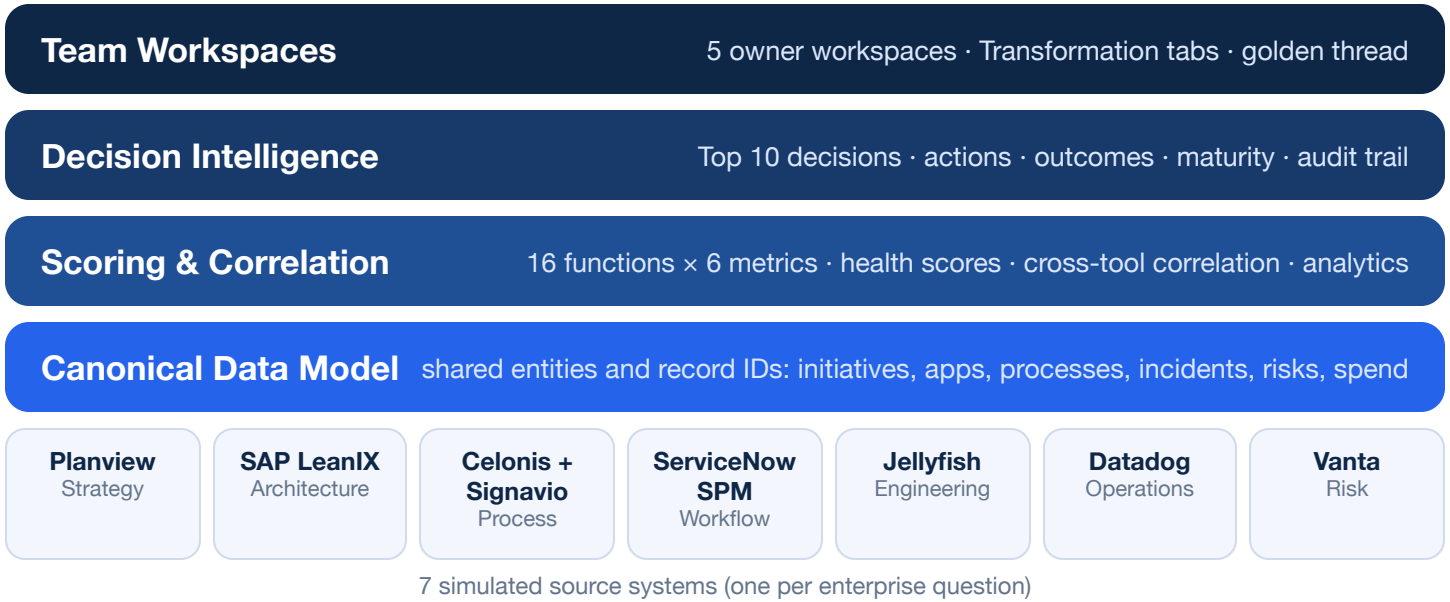
GOVERNANCE · TEDIF

AI Recommends. Humans Decide.

Trusted Enterprise Decision Intelligence Framework: AI gathers signals and recommends, a named person decides at a gate, and every step is recorded in an audit trail.

Architecture: 3 Business Units × 7 Sources × 16 Functions

A layered design: sources at the bottom, a common data model, scoring and correlation, then decisions and team workspaces on top.

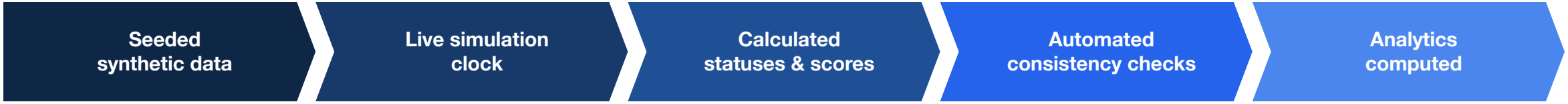


- **Build once, configure per industry:** Meridian Bank, Arvant Industries and Orbit Retail run on the same engine.
- **7 source systems:** each represents a real enterprise tool category and feeds the canonical model.
- **16 enterprise functions:** from Technology and Data & AI to Finance, HR, Legal and Procurement, each with 6 scored metrics.
- **Shared record IDs** are what make cross-tool correlation, evidence and drill-down possible.

03 · HOW IT WORKS

A Realistic Simulation, Built to Be Trusted

The simulation behaves like a real enterprise: data, events and calculations are consistent, repeatable and checked automatically.



- **Seeded data:** the same realistic enterprise is generated every time, so the demo is repeatable.
- **Live clock:** the simulation runs in real time; events such as a new regulatory circular or a breach clock arrive on stage.
- **Nothing is typed by hand:** every status, score, risk level and decision priority is calculated from the records.
- **Automated checks:** every build validates 3 domains plus team cross-checks, for example spend ledgers reconcile and every obligation maps to a real control.

Analytics Actually Computed

Holt-Winters forecasting · EWMA anomaly detection · Monte Carlo (P10 / P50 / P90) · NPV · IRR · payback · S-curve adoption · PSI model drift · Pareto 80/20 · Queueing (M/M/1) · SRE error budget · Glide path & ETA

Honest framing: this is synthetic data, not client data. The design takes real tool exports as the next step.

Real Data Reality Check: Six Public Datasets

Public, credited datasets beside the simulation prove the patterns are real: end of life, exploited vulnerabilities, regulatory volume, drift, Pareto and cost allocation.

RAM

REAL DATA · endoflife.date

12 of 20

matched technology versions are already past real vendor end of life; the simulation was optimistic for 6

VAIBHAV

REAL DATA · RBI notifications

22

real RBI notifications in two weeks (23 Sep to 7 Oct 2026): the true intake volume to triage

VAIBHAV

REAL DATA · Kaggle card fraud

PSI 2.42

on 284,807 real transactions: stable day to day (0.19), sharp drift in fraud-wave hours

PANKAJ

REAL DATA · UCI ServiceNow log

79.2%

of 24,918 real incidents come from 20% of categories; only 1.5% linked to a problem record

SANTHOSH

REAL DATA · FOCUS cloud billing

62.7%

of real cloud cost is fully allocatable; 37% cannot be traced to an application

RAM

REAL DATA · CISA KEV catalogue

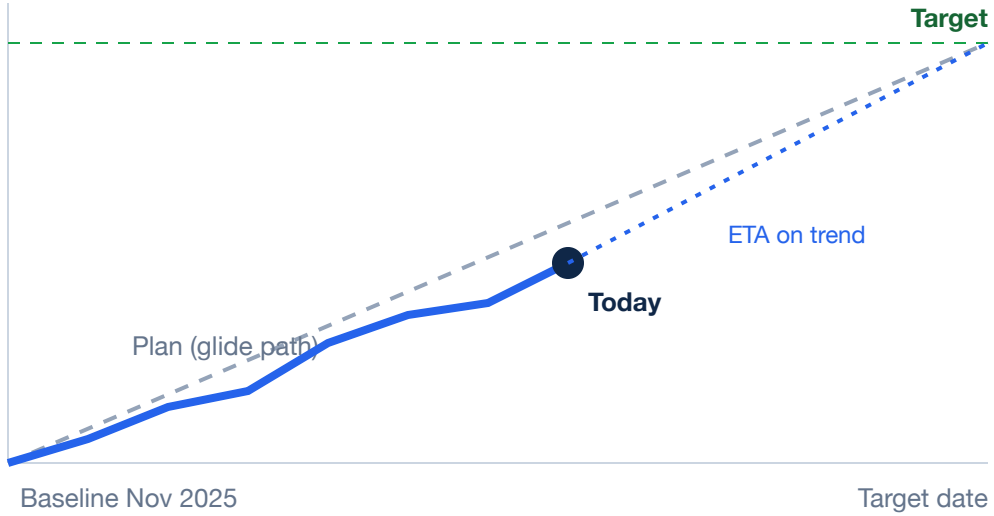
1,734

vulnerabilities exploited in the wild; 103 added in 90 days, 361 used by ransomware

- **Simulation for the enterprise, real data where it matters.** Each panel in the app is labelled REAL DATA with source, licence and fetch date (snapshot 8 Oct 2026).
- **Scores stay simulated:** real data is a reality check beside the simulation, never mixed into it; no personal data is used.

How We Measure Transformation

Every owner moves from old state to target state. Progress is compared with time used, so we know whether we are transforming fast enough.



- **6 measures per owner, 30 in total,** each with an old state (baseline Nov 2025), today, and a target with a date.
- **Progress** = share of the gap from baseline to target already closed.
- **Time used** = share of the time from baseline to the target date already passed.
- **Glide path:** comparing progress with time used gives Ahead, On Track or Behind, so a slow start is visible early.
- **ETA:** projected from the last quarter's trend; flagged when the trend is flat or moving away.
- **Same rule for everyone,** so a team score (about 59% today) can be compared across owners.

Ahead
progress ≥ time used + 10

On Track
within ±10 points

Behind
progress < time used – 10

04 · LIVE DEMO

Live Demo: Five Owners, One Connected Decision

Live site: cto360.vercel.app. The scenario plays on a simulation clock while we present.

0	1	2	3	4	5	6
Ram /team	**Ram** Control Tower	**Suman** AI Transformation	**Vaibhav** Regulatory & AI Gov.	**Santhosh** Spend & Investment	**Pankaj** ERP RCA & Capacity	**Ram** /team
Problem statement, team scorecard, golden thread starts live	5 agents, correlation, approve Digital Payments in the Decision Center	Funnel, promote a pilot, Monte Carlo, onboard "Insurance"	Live circular (+50 s), model drift, breach clock (+80 s)	TBM Sankey, festive cloud spike, NPV / IRR	Pareto + 5 Whys, capacity gauge, festive slider	Golden thread complete: old state to target state, measurably

- **Before starting:** open the Team page and press Reset in the header so the scenario replays from the beginning.
- **Each presenter, about 3 minutes:** Transformation tab, two functional tabs with one live action, then the Frameworks Applied tab.

Ram: CTO Control Tower

INTEGRATE, DECIDE, PROVE OUTCOMES **BP1:** Multi-domain framework; decision intelligence and agents

OLD STATE

Signals scattered across 7 tools and 3 business units; decisions take weeks and outcomes are rarely measured.

▼

TARGET STATE

One control tower: correlated signals become owned decisions within days, with outcomes tracked against targets.

Signal-to-decision time 83% of the gap closed

Old state 21 days **Now 6 days** Target 3 days

Decisions with measured outcome 28% of the gap closed

Old state 0% **Now 22%** Target 80%

Business units on one framework 100% of the gap closed

Old state 0 **Now 3** Target 3

What We Built

- **Multi-domain view:** two heatmaps compare all business units and functions on one scale.
- **Agents & correlation:** 5 simulated agents feed a correlation engine; humans approve decisions.
- **Golden thread:** one cross-team scenario plays live on the simulation clock.
- **Programmes:** status by business unit and the not-on-track list.

Frameworks Applied With Metrics

Balanced Scorecard | Cynefin | RAPID | OODA loop

Named decider on every decision 30% → 100% (RAPID); average open-decision age 19 → 6.4 days, target 3.

Suman: AI Transformation

FROM PILOTS TO MEASURABLE VALUE, FOR ANY ORGANISATION **BP1:** Identify ROI; transform any organisation to AI

OLD STATE

AI pilots everywhere, few in production; ROI guessed after the money is spent.

▼

TARGET STATE

A governed AI portfolio where every use case has ROI identified before investment and pilots are promoted through gates.

Pilot-to-production conversion — 55% of the gap closed

Old state 18% | **Now 41%** | Target 60%

Average AI payback — 50% of the gap closed

Old state 30 mo | **Now 24 mo** | Target 18 mo

Live use cases past the chasm — 79% of the gap closed

Old state 18% | **Now 63%** | Target 75%

What We Built

- **Readiness:** AI maturity radar per business unit.
- **Portfolio & gates:** 38 AI use cases, funnel, value versus feasibility, gate review with promote.
- **ROI & benefits:** NPV, payback and Monte Carlo P10 to P90 range.
- **Onboard any organisation:** 6 industry templates generate a plan and a Business Model Canvas.

Frameworks Applied With Metrics

Three Horizons | Diffusion of Innovations | Crossing the Chasm | Business Model Canvas | Stage-Gate | Value vs Effort

Three Horizons mix H1 38% / H2 40% / H3 22% versus a 70 / 20 / 10 guide shows "pilot purgatory": too little in live use cases.

Vaibhav: Regulatory & AI Governance

COMPLIANCE AT THE SPEED OF CHANGE; TRUSTWORTHY AI AND DATA **BP1:** Automate regulatory compliance (Banking); DPDP and AI governance

OLD STATE

Regulatory circulars mapped by hand over weeks; AI models and personal data not consistently governed.

▼

TARGET STATE

Circular to obligation to control to evidence in days, every model registered and monitored, DPDP ready.

Breach reports on time — 60% of the gap closed

Old state 50% | **Now 80%** | Target 100%

Risks covered by all three lines — 51% of the gap closed

Old state 20% | **Now 61%** | Target 100%

What We Built

- **Regulatory change:** a live circular arrives, obligations are mapped to controls and approved.
- **AI model register:** PSI drift monitoring; the fraud model at PSI 0.37 is restricted.
- **Data governance & DPDP:** consent coverage, lineage and a live breach clock.
- **Audit readiness:** evidence per control for internal audit and the regulator.

Frameworks Applied With Metrics

Three Lines Model | Risk process | Risk heat map | NIST AI RMF (spirit)

Three Lines applied to 33 high and critical risks: first line owns, second oversees, third assures.

Santhosh: Technology Spend & Investment

EVERY RUPEE TRACEABLE TO A SERVICE AND A VALUE **BP1:** Governance system for investment; CAPEX and OPEX

OLD STATE

Technology spend in cost pools nobody can trace; cloud waste; investments approved without benefit tracking.

▼

TARGET STATE

Spend allocated to services, cloud under FinOps control, and every investment value-tracked after go-live.

Spend allocated to services (TBM) 58% of the gap closed

Old state 40% **Now 72%** Target 95%

Cloud waste (idle / unused) 55% of the gap closed

Old state 28% **Now 17%** Target 8%

Investments value-tracked 51% of the gap closed

Old state 15% **Now 58%** Target 100%

What We Built

- **CAPEX / OPEX & TBM:** a Sankey from cost pools to towers to services.
- **Budget & forecast:** Holt-Winters forecast and EWMA anomalies, for example the festive cloud spike.
- **FinOps & licences:** rightsizing and licence utilisation.
- **Investment governance:** NPV, IRR, payback; security programmes judged on risk reduced.

Frameworks Applied With Metrics

Run-Grow-Transform | TCO | Sensitivity (tornado) | NPV / IRR | TBM | FinOps

Run-Grow-Transform gap from the 60 / 25 / 15 target mix reduced from 30 to 14.2 points; "Grow" is under-funded.

Pankaj: ERP RCA & Capacity

FIX CAUSES, NOT SYMPTOMS; CAPACITY BEFORE THE PEAK **BP1:** ERP root-cause analysis; infrastructure capacity; production automation

OLD STATE

The same ERP incidents recur; capacity is added after the peak hurts; production steps rely on manual hand-offs.

▼

TARGET STATE

Root causes fixed permanently, capacity headroom ready before month-end and festive peaks, production steps automated.

Recurring ERP problems — 55% of the gap closed

Old state 14 | **Now 8** | Target 3

Time to permanent fix — 60% of the gap closed

Old state 45 days | **Now 24 days** | Target 10 days

Minimum capacity headroom — 41% of the gap closed

Old state 8% | **Now 17%** | Target 30%

What We Built

- **ERP RCA:** Pareto of incident causes and a 5 Whys chain to the root cause.
- **Capacity & forecasting:** Holt-Winters forecast of utilisation against the limit.
- **Peak scenarios:** festive slider; 0% extra capacity gives 1,546 ms, 25% brings latency within target.
- **Production automation:** automated steps tracked with a Lewin change strip.

Frameworks Applied With Metrics

Theory of Constraints | Pareto 80/20 | 5 Whys | ITIL problem mgmt | SRE error budget | Lewin

Theory of Constraints names the constraint per peak (for example month-end: IIoT ingestion CPU) and the next one after it.

04 · LIVE DEMO

The Golden Thread: One Connected Decision

Six steps play live on the simulation clock, from first signal to approved decision and recovered capacity.

1

Pankaj +15 s

Month-end batch overrun; capacity headroom below 10%

Source: Datadog · ServiceNow

2

Ram +40 s

Control tower correlates it with festive and salary-day peaks

Source: Correlation engine

3

Santhosh +65 s

Options costed: cloud burst vs hardware vs re-schedule

Source: Cloud billing · Planview

4

Suman +90 s

AI demand forecasting proposed to predict peaks

Source: AI portfolio · MLOps

5

Vaibhav +115 s

Governance check: data residency and model risk

Source: Vanta · model register

6

Ram +140 s

Decision approved; actions issued; headroom recovers

Source: Decision Center

- **Scenario: "Month-End Close Meets Festive Peak".** Every step cites real simulated records, so the audience can click through to the evidence.
- **The point:** no single owner or tool could solve this alone; CTO360 connects five owners into one decision with a measured outcome.

Frameworks Applied, Not Just Listed

"You do not need 100 frameworks." Choose the right one when a CEO, CFO or COO brings a problem, and measure it.

16 **Frameworks applied with tracked metrics**
Balanced Scorecard, Cynefin, RAPID, OODA, Three Horizons, Diffusion, Chasm, BMC, Three Lines, Run-Grow-Transform, TCO, Sensitivity, Theory of Constraints, Lewin, Kotter, ADKAR

15 **Frameworks built into the screens earlier**
Stage-Gate, Value vs Effort, NPV / IRR / ROI, maturity models, risk process and heat map, RACI, ITIL, 5 Whys, Pareto, SRE, TBM, FinOps, TEDIF

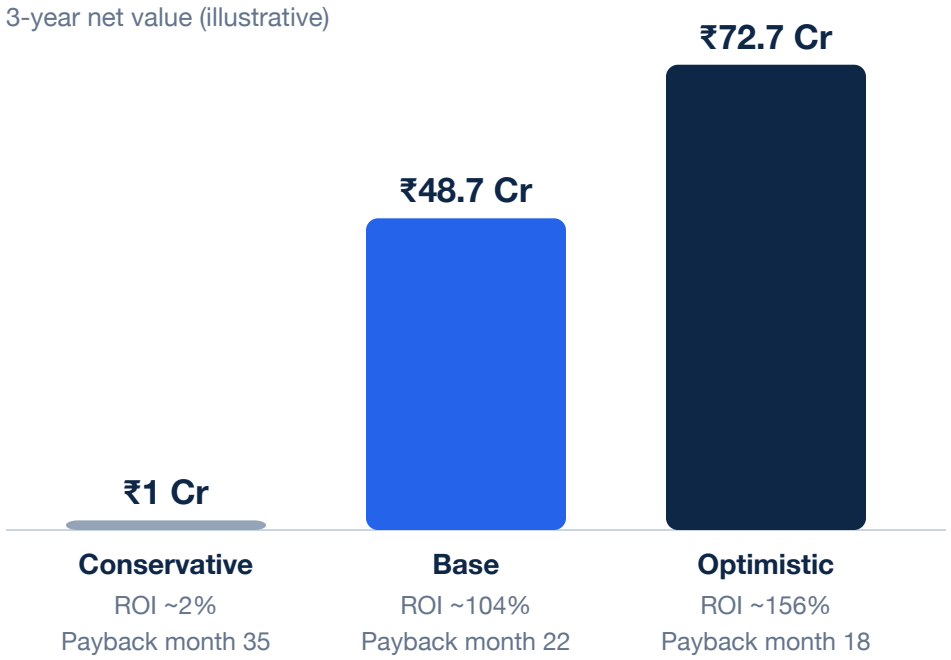
10 **Analytics techniques computed live**
Holt-Winters, EWMA, Monte Carlo, NPV / IRR, S-curve, PSI drift, Pareto, queueing, error budget, glide path

69 **Reference library cards**
Every framework, technique and concept explained with a small diagram and the business question it answers

- **Lean by design:** a framework is applied only where it serves an owner's business problem and produces a tracked metric; the rest stays in a reference library.
- **Team level change:** Kotter 8 steps turn green as decisions are approved in the app; ADKAR adoption score 1.8 → 3.2 out of 5 (target 4).

Value Case: Commit to the Conservative Case

A realistic, defensible value case: modest commitment, credible upside, and risk avoidance on top.



- **Investment:** about ₹30.7 Cr to build over 2 years, plus 10 to 20% a year to run.
- **Value ramp:** 15% of target value in year 1, 45% in year 2, 75% in year 3.
- **We commit to the conservative case** and show the base case as the expectation; payback around month 22 in the base case.
- **Risk avoidance is extra:** DPDP penalties for failed safeguards can reach about ₹250 Cr; one avoided incident can fund the programme.
- **Operational gains already modelled:** regulatory mapping 21 → 4 days, manufacturing RCA 38 → 9 hours, CAPEX approval 45 → 21 days.

Why CTO360 Is Different

Point tools are essential, but each answers its own question. CTO360 answers the CTO's question: what matters most, and what should we decide?

Capability	Point tools (Planview, LeanIX, ServiceNow, Datadog, …)	CTO360
Shows what is happening inside one function	✔ Yes	✔ Yes
Connects signals across tools into one story	✖ No	✔ Yes
Prioritised Top 10 decisions with owner and deadline	✖ No	✔ Yes
Tracks decision to action to measured outcome	◐ Partial	✔ Yes
One comparable view across business units	✖ No	✔ Yes
Governed AI with named human decider and audit trail	◐ Varies	✔ Yes
New industry added by configuration	✖ No	✔ Yes

- **We do not replace existing tools; we connect them.** CTO360 is the decision layer a CTO needs on top of the tools the enterprise already pays for.

Roadmap: From Simulation to Production

The blueprint is ready. The next step is real data and live AI, under the same governance.

Today
Capstone build (v1.0)

6 to 12 Months
Phase 2 and pilot

2 to 3 Years
Scaled platform

Today

- Working simulation of 3 enterprises and 7 sources
- 5 live owner workspaces with 30 measures
- Six real public datasets as reality checks
- Automated consistency checks on every build

6 to 12 Months

- Live AI call with RAG on 3 decisions
- Real baselines from one business unit
- Real tool exports replace simulated sources
- Persistent audit trail and a governed pilot

2 to 3 Years

- Platform in 2 to 3 business units
- ₹25 to 50 Cr a year if adoption holds
- Compliance readiness (DPDP, EU AI Act)
- New industries added by configuration

Guardrail from TEDIF: no production AI before day 90; every phase passes a human-approved gate.

IN ONE LINE

Five Owners. One Connected Decision. Measured Outcomes.

AI Recommends. Humans Decide.

Enterprise tools tell leaders what is happening inside each function. CTO360 connects those signals to show the CTO what matters, where to intervene, what decision to make, and whether it worked.

Thank you · Questions?

Live: cto360.vercel.app

APPENDIX A1

The 16 Enterprise Functions

Every function is scored the same way in every business unit, which makes the heatmap comparable.

Technology Owner: CTO / CIO	**Enterprise Architecture** Owner: CTO / CIO	**Engineering & R&D** Owner: CTO	**Data & AI** Owner: CDO / CAIO
Cybersecurity Owner: CISO	**Product & Innovation** Owner: CPO / CTO	**Corporate Strategy** Owner: CEO / CSO	**Finance** Owner: CFO
Operations Owner: COO	**Sales** Owner: CRO	**Marketing** Owner: CMO	**Customer Experience & Service** Owner: CXO / COO
Human Resources Owner: CHRO	**Risk & Compliance** Owner: CRO	**Legal** Owner: CLO	**Procurement & Vendor Management** Owner: CPO / COO

Shaded: 6 technology functions owned by the CTO organisation. Unshaded: 10 business functions where technology shows up as a signal. Each function has 6 metrics: strategy, performance, cost, technology, risk and transformation.

APPENDIX A2

The 7 Source Systems and the Question Each Answers

Each source is simulated with realistic records and linked to the others through shared record IDs.

Simulated source	Area	The question it answers
Planview	Strategy & Portfolio	Are investments aligned to strategy and on track?
SAP LeanIX	Enterprise Architecture	Is the application estate fit for the future?
Celonis + SAP Signavio	Process Intelligence	Where does the process lose time or money?
ServiceNow SPM	Portfolio & Workflow	Is demand flowing through to delivery?
Jellyfish	Engineering Intelligence	Is engineering capacity on the right work?
Datadog	Operations & Observability	Are critical services healthy and fast?
Vanta	Security, Risk & Governance	Are controls effective and risks within appetite?

Product names are used only to describe tool categories. CTO360 is an independent demonstration on synthetic data and is not affiliated with or endorsed by these vendors.

APPENDIX A3

Anticipated Questions

Short, honest answers to the questions we expect from the panel.

Is this real data?

The enterprise is simulated and every score is calculated and checked. Beside it, six real public datasets (RBI, CISA, Kaggle, UCI, FOCUS, endoflife.date) act as labelled reality checks.

Why not just buy Planview or ServiceNow?

We do not replace them, we connect them. None correlates across tools into a prioritised decision with an owner and a measured outcome.

Where is the AI?

Simulated agents, correlation and recommendations today (rule-based and labelled). Next: a live LLM call with RAG on 3 decisions.

How do you stop AI from making bad decisions?

AI only recommends. A named decider approves at a gate (RAPID), models are monitored for drift (PSI), and the Three Lines Model provides oversight.

How does it work for a new industry?

Configuration, not rebuild. Retail was added by configuration only; the Onboard wizard creates a plan for any industry.

What does it cost and when does it pay back?

About ₹30.7 Cr build; base payback around month 22; we commit to the conservative case and treat risk avoidance as upside.

APPENDIX A4

Disclaimer

Full disclaimer: cto360.vercel.app/about

- CTO360 is an independent capstone demonstration built on **synthetic, simulated data**, with six **credited public datasets** shown as labelled reality checks.
- All organisations (Meridian Bank, Arvant Industries, Orbit Retail), people, figures and events are illustrative.
- Third-party product names are referenced solely to describe tool categories and conceptual integration scenarios. CTO360 is **not affiliated with or endorsed by** the referenced vendors.
- Financial figures, ROI and payback are modelled estimates for discussion, not forecasts or advice.
- Framework applications are simplified for teaching purposes; they are not certifications or audits.